

Bandwidth Estimation: Metrics, Estimation Techniques and Tools

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Overview of QoS parameters

▶ Delay-related metrics

- ▶ Round-trip time
- ▶ One way delay
- ▶ Jitter
- ▶ Queuing delay
- ▶ Latency (packet size/bandwidth)
- ▶ Propagation delay

▶ Bandwidth-related parameters

- ▶ Capacity
- ▶ Available bandwidth
- ▶ Achievable throughput
- ▶ Bulk Transfer Capacity (BTC)

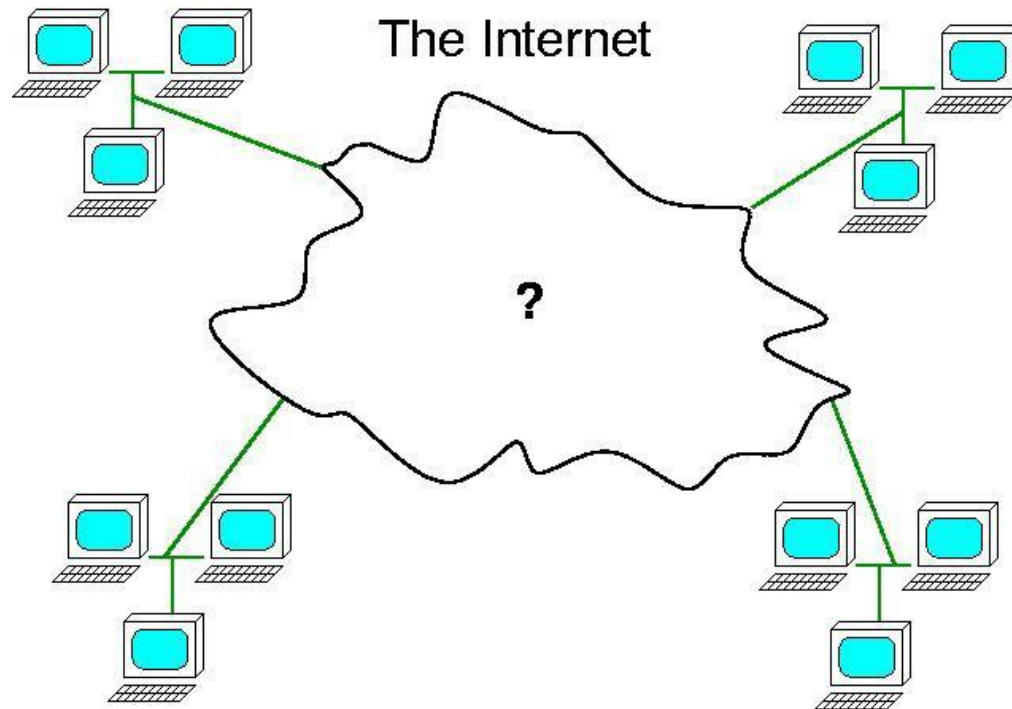
▶ Packet-related metrics

- ▶ Packet loss
- ▶ Packet reordering
- ▶ Packet corruption
- ▶ Packet size distribution
- ▶ Packet interarrival time distribution

Bandwidth-related parameters

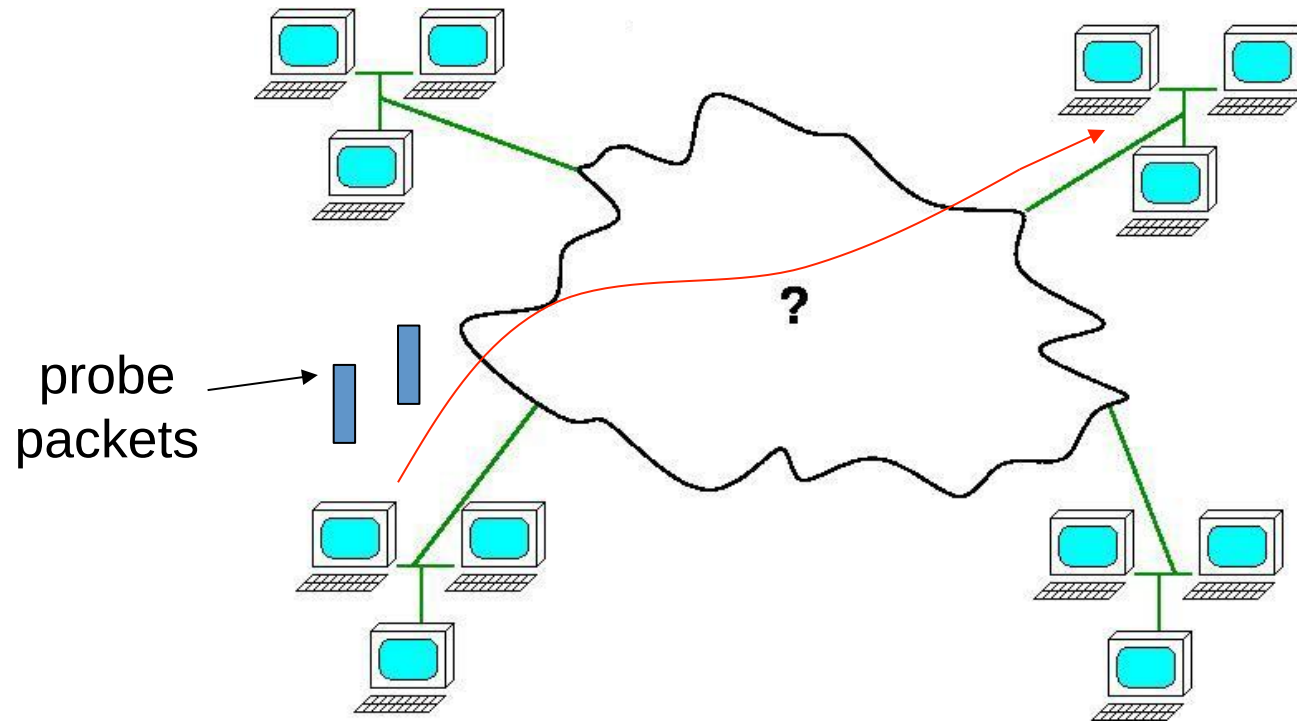
- Motivation
- Metrics
 - Capacity
 - Available bandwidth
 - Achievable throughput
 - Bulk transfer capacity
- Bandwidth estimation techniques and tools
- Taxonomy of tools

Internet as a “black box”



- Several network properties are important for applications and transport protocols:
 - Delay, loss rate, capacity, congestion, load, etc
- But routers and switches do not provide direct feedback to end--hosts (except ICMP, also of limited use)
 - Mostly due to scalability, policy, and simplicity reasons

Can we guess what is in the black box?



- End-systems can infer network state through end-to-end (e2e) measurements
 - Without any feedback from routers
 - Objectives: accuracy, speed, non-intrusiveness

Motivation

- Bandwidth has been a critical and precious resource in various kinds of networks.
- Having a good estimate of bandwidth is, for example, important for network error detection and diagnosis as well as for efficient operation of bandwidth dependent/data-intensive Internet applications.
- Popular examples where knowledge of bandwidth can be valuable are
 - validation of service level agreements,
 - detection of congested or underutilized links,
 - capacity planning support,
 - admission control policies at massively-accessed content servers,
 - network tomography for tracking and visualizing Internet topology,
 - dynamic server selection for downloads,
 - optimized congestion control for reliable transport protocols (e.g. for TCP) and
 - optimized network route selection.

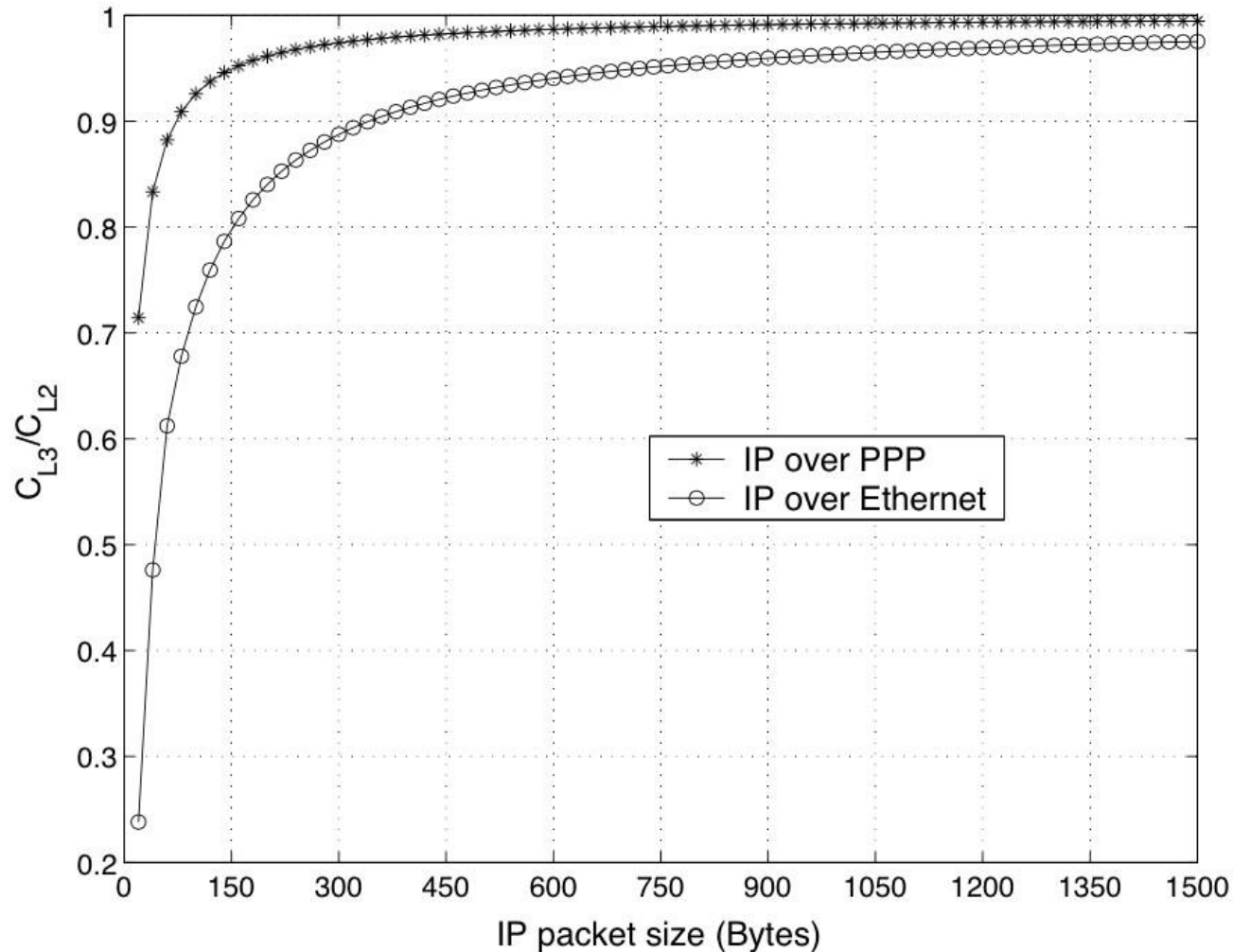
Hops, segments, and paths

- Segment: a link at layer 2
 - Physical point-to-point link
 - Virtual circuit
 - Shared access LAN (ETHERNET, FDDI)
- Hop: a link at layer 3
 - Sequence of segments connected by switches, bridges and other layer 2 devices. Hops are only separated by routers.
- Path p from s to v
 - Sequence of hops from s to v

Capacity

- At layer 2 link (segment):
 - Transmission rate limited by:
 - capacity of the physical link,
 - speed of the transmitter/receiver hardware
- At IP layer:
 - Lower rate due to overheads in layer 2 by encapsulation and framing

Effect of packet size on capacity usage



Capacity (cont.)

- Capacity of a hop
 - Maximum possible IP layer transfer rate at that hop
 - Maximum layer 2 transfer can occur only with Maximum Transmission Unit sized packets
 - Bit rate measured at IP layer transferring MTU sized packets
- Capacity of a path:
 - Minimum link capacity determines capacity of the path

$$C = \min_{i=1,2,\dots,H} C_i$$

Problems with estimating capacity

- Traffic shapers
- Rate limiters
- Wireless networks like 802.11 operate at different rates
 - 802.11g: 54, 11, 5.5, 2 or 1 Mbps
- Definition holds during time at which the capacity remains constant

Available bandwidth (1)

- The available bandwidth of a link relates to the unused, or “spare”, capacity of the link during a certain time period
- At any time
 - Link used fully → utilization =1
 - Not used → utilization =0
- Average utilization from time $t-\alpha$ to t is given by

$$\bar{u}(t-\alpha, t) = \frac{1}{\alpha} \int_{t-\alpha}^t u(x) dx$$

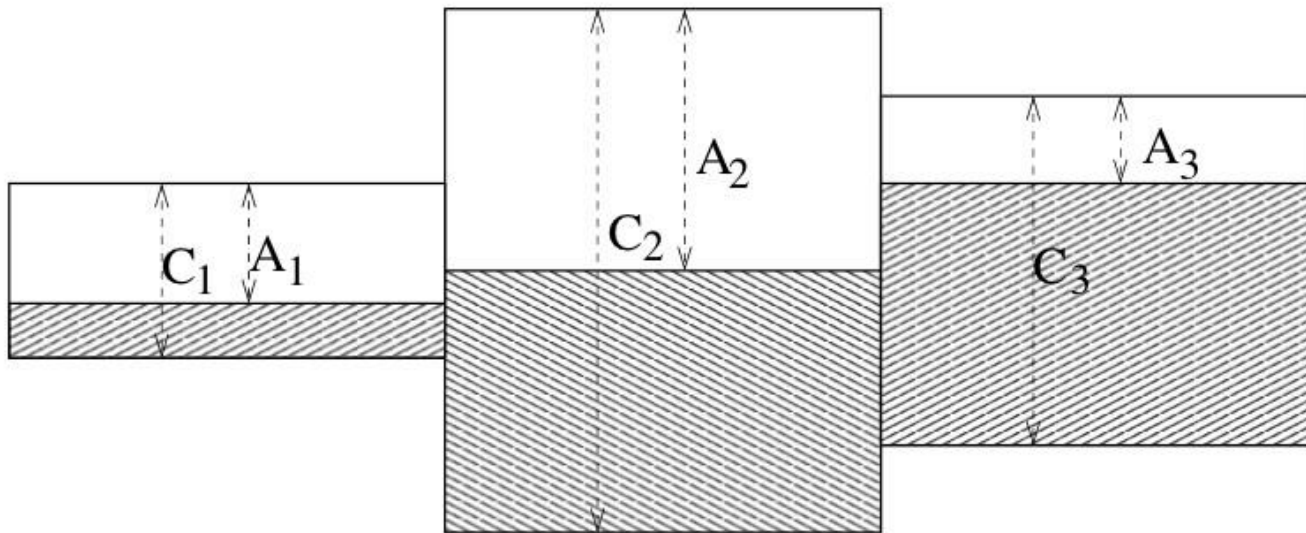
- Then the available bandwidth during that period:

$$A_i = (1 - u_i) C_i$$

Available bandwidth (2)

- Available bandwidth along the path

$$A = \min_{i=1,2,\dots,H} A_i$$



Assumptions

- Link utilization remains constant over the duration of measurement
 - Reasonable for short intervals
 - Load variations impact the measurement over a long period
- So available bandwidth measurements should be done quickly
 - Since capacity remains constant, those measurements need not be made quickly

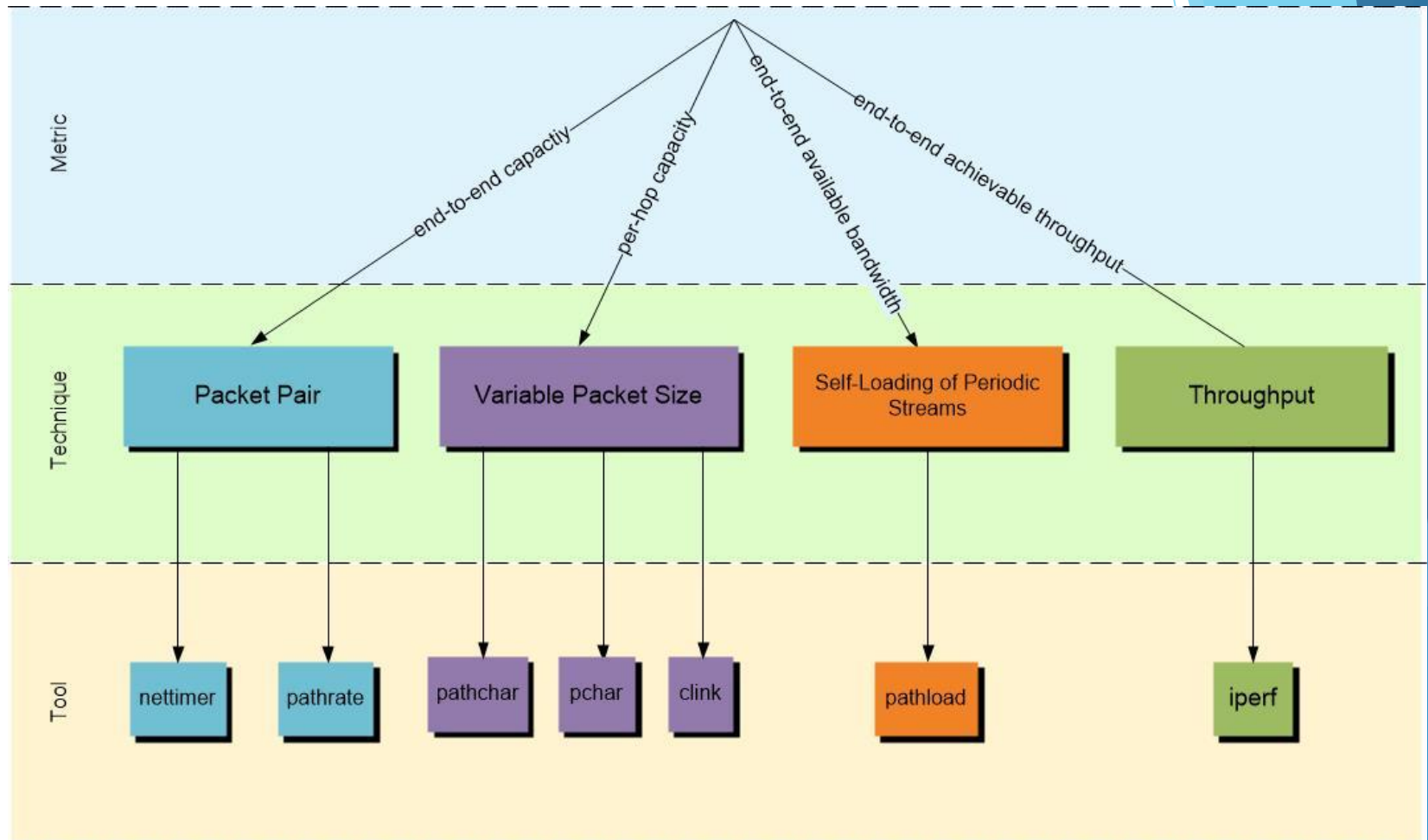
Achievable throughput and Bulk Transfer Capacity

- Achievable throughput depends on various parameters
 - Transport protocol, i.e. UDP or TCP
 - Congestion window
 - RTT
 - Slow start mechanism
 - Concurrently established connections
 - Capacity and load along the path
 - CPU, System IO through, memory limitations etc.
- BTC
 - Maximum capacity obtainable by a single TCP connection

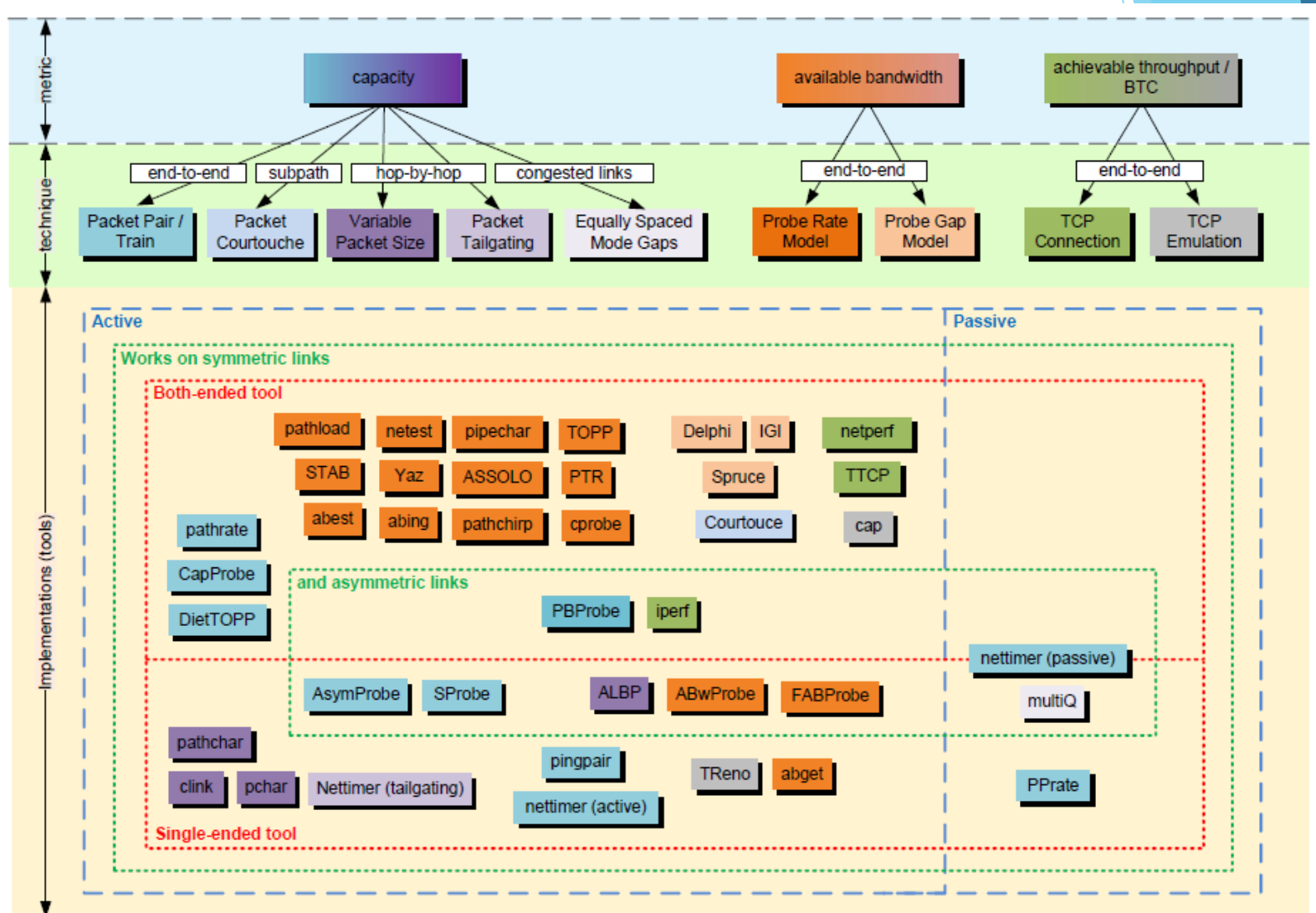
Differences between BTC and available bandwidth

- BTC is TCP specific
 - Available bandwidth is transport protocol independent
- BTC depends on the how a TCP connection throughput is affected by other flows
 - Available bandwidth assumes average load remains constant and estimates additional bandwidth

An overview: metrics, techniques and tools



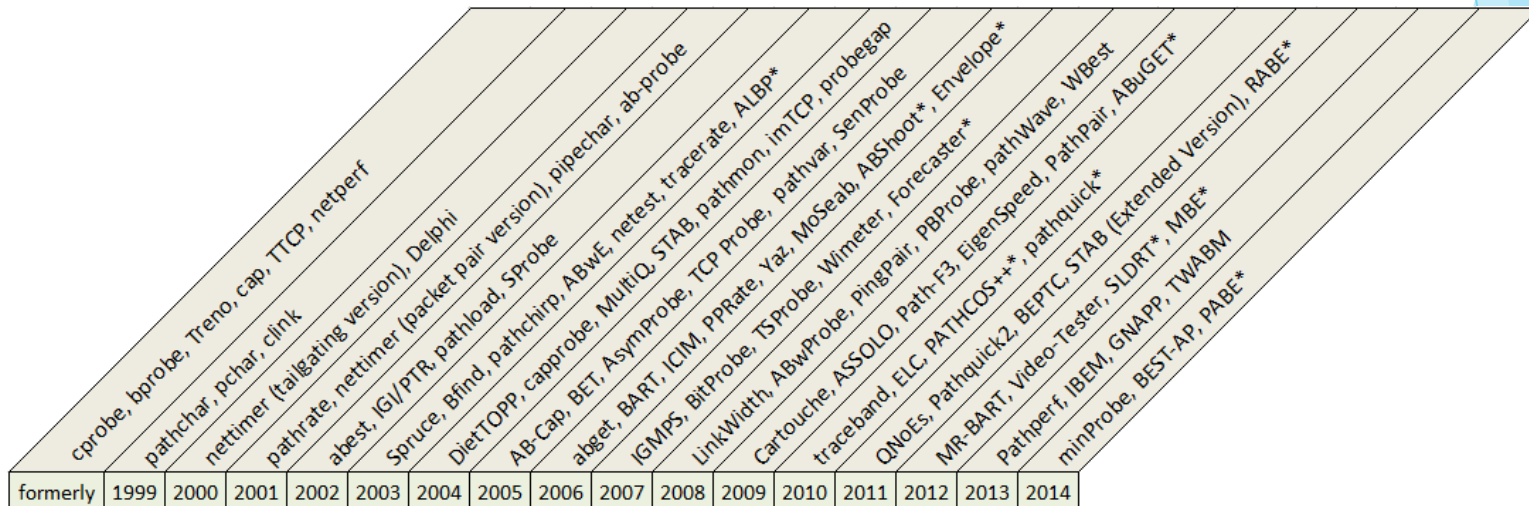
An overview: metrics, techniques and tools - Extended



An overview: metrics, techniques and tools.

Taxonomy of tools (1)

- Estimation tools show a wide spectrum of different assumptions, characteristics, advantages and limitations
 - Active vs. passive estimation tools
 - Intrusive vs. lightweight estimation tools
 - Single-ended vs. double-ended estimation tools
 - Offline vs. online (or real-time) estimation tools
 - Ability to measure asymmetric, wireless or high-speed links
 - Different levels of achievable estimation accuracy
 - Differences in estimation time needed and probing overhead caused



Evolution of bandwidth estimation tools. Each tool is assigned to its publication date.

Tools marked with an asterisk (*) are implemented in simulation only.

Taxonomy of tools (2)

- From the point of a user's view, the wishing list for an ideal estimation technique and tool is long and manifold, and includes requirements such as
 - accurate, consistent and reliable estimates,
 - low overhead,
 - fast and robust estimation procedures,
 - resilience to both cross traffic and its rapid changing conditions,
 - independence of the measurement end-host performance and system capabilities and
 - applicability to mixed paths consisting of wired, high-speed and/or wireless links.

Thank you for your attention!